

Homework 1

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Problem 1

(a) I scale u first and then subtract v one component at a time.

$$2u = [4, -2, 6]^T, \text{ so } 2u - v = [4 - 1, -2 - 4, 6 - (-2)]^T = [3, -6, 8]^T.$$

Now the dot product with w , multiplying matching entries and adding them up:

$$(3)(-4) + (-6)(2) + (8)(6) = -12 - 12 + 48 = 24$$

$$\text{Final answer: } (2u - v) \cdot w = 24$$

(b) Half of w is $(1/2)w = [-2, 1, 3]^T$, and adding it to v entry by entry gives

$$v + (1/2)w = [1 + (-2), 4 + 1, -2 + 3]^T = [-1, 5, 1]^T.$$

The norm is the square root of the sum of the squares of the entries, so

$$\|v + (1/2)w\| = \sqrt{(-1)^2 + 5^2 + 1^2} = \sqrt{1 + 25 + 1} = \sqrt{27}$$

$$\text{Final answer: } \|v + (1/2)w\| = \sqrt{27}$$

(c) The angle comes from $\cos \theta = (u \cdot w) / (\|u\| \|w\|)$, so I need the dot product of the

two vectors and both of their lengths.

$$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 12$$

$$\|u\| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{4 + 1 + 9} = \sqrt{14}$$

$$\|w\| = \sqrt{(-4)^2 + 2^2 + 6^2} = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

The lengths multiply to $\sqrt{14} \cdot 2\sqrt{14} = 2 \cdot 14 = 28$, so $\cos \theta = 12/28 = 3/7$.

Taking the inverse cosine, $\theta = \arccos(3/7) = 64.6231\dots^\circ$, which rounds to 64.62° .

$$\text{Final answer: } \theta \approx 64.62^\circ$$

Problem 2

(a) A is 3×3 and B is 2×3 . For AB we would need A 's 3 columns to match B 's 2 rows, and

$3 \neq 2$, so AB is undefined. BA is fine: $(2 \times 3)(3 \times 3)$ comes out 2×3 .

$$\text{row 1: } 1(2) + 0(1) + (-2)(0) = 2, \quad 1(-1) + 0(3) + (-2)(4) = -9, \quad 1(0) + 0(-2) + (-2)(1) = -2$$

$$\text{row 2: } 3(2) + 1(1) + 1(0) = 7, \quad 3(-1) + 1(3) + 1(4) = 2, \quad 3(0) + 1(-2) + 1(1) = -1$$

$$\text{Final answer: } BA = \begin{bmatrix} 2 & -9 & -2 \\ 7 & 2 & -1 \end{bmatrix}$$

(b) A is 3×3 and v is 3×1 , so Av is defined and 3×1 , each entry a row of A dotted with v .

$$Av = [2(1) + (-1)(-2) + 0(3), \quad 1(1) + 3(-2) + (-2)(3), \quad 0(1) + 4(-2) + 1(3)]^T = [4, -11, -5]^T$$

vA is undefined, because v has only 1 column while A has 3 rows, and $1 \neq 3$.

$$\text{Final answer: } Av = [4, -11, -5]^T, \quad vA \text{ undefined}$$

(c) Aw is undefined: A has 3 columns but the row vector w has only 1 row, so $3 \neq 1$.

wA is $(1 \times 3)(3 \times 3)$, so it is a 1×3 row vector:

$$[2(2) + 0(1) + (-1)(0), \quad 2(-1) + 0(3) + (-1)(4), \quad 2(0) + 0(-2) + (-1)(1)] = [4 \quad -6 \quad -1]$$

(d) vw is $(3 \times 1)(1 \times 3)$, so it is 3×3 and each entry is an entry of v times an entry of w :

$$vw = \begin{bmatrix} 2 & 0 & -1 \\ -4 & 0 & 2 \\ 6 & 0 & -3 \end{bmatrix}$$

wv is $(1 \times 3)(3 \times 1) = 1 \times 1$, which is just the dot product of w and v :

$$wv = 2(1) + 0(-2) + (-1)(3) = 2 - 3 = -1, \text{ so } wv = [-1].$$

(e) A is square, so $A^2 = AA$ is defined and is 3×3 . Going row by row:

$$\text{row 1: } [2(2) + (-1)(1) + 0(0), 2(-1) + (-1)(3) + 0(4), 2(0) + (-1)(-2) + 0(1)] = [3, -5, 2]$$

$$\text{row 2: } [1(2) + 3(1) + (-2)(0), 1(-1) + 3(3) + (-2)(4), 1(0) + 3(-2) + (-2)(1)] = [5, 0, -8]$$

$$\text{row 3: } [0(2) + 4(1) + 1(0), 0(-1) + 4(3) + 1(4), 0(0) + 4(-2) + 1(1)] = [4, 16, -7]$$

$$\text{Final answer: } A^2 = \begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 16 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 0 & -8 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 16 & -7 \end{bmatrix}$$

(f) $B^2 = BB$ is undefined. B is 2×3 , so squaring it would need its 3 columns to match its

own 2 rows. Only a square matrix can be multiplied by itself.

Problem 3

(a) I write the augmented matrix and clear the first column first.

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 2 & 3 & 1 & | & 2 \\ -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 + R_1: \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$R_2 \rightarrow -R_2: \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2: \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 0 & 10 & | & 10 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 10 & | & 10 \end{bmatrix}$$

$$R_3 \rightarrow (1/10)R_3: \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 0 & 1 & | & 1 \end{bmatrix}$$

$$[0 \ 1 \ -3 \mid -4]$$

$$[0 \ 0 \ 1 \mid 1]$$

Back substituting from the bottom up: $x_3 = 1$, then $x_2 - 3(1) = -4$ gives $x_2 = -1$, then

$x_1 + 2(-1) - 1 = -1$ gives $x_1 = 2$. Every column has a pivot, so there are no free variables and this is the only solution.

Final answer: $x = [2, -1, 1]^T$

(b) $[1 \ 2 \ -1 \mid 4]$

$$[2 \ 5 \ 1 \mid 6]$$

$$[3 \ 7 \ 0 \mid 10]$$

$$R_2 \rightarrow R_2 - 2R_1: [1 \ 2 \ -1 \mid 4]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[3 \ 7 \ 0 \mid 10]$$

$$R_3 \rightarrow R_3 - 3R_1: [1 \ 2 \ -1 \mid 4]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[0 \ 1 \ 3 \mid 0]$$

$$R_3 \rightarrow R_3 - R_2: [1 \ 2 \ -1 \mid 4]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[0 \ 0 \ 0 \mid 2]$$

That bottom row says $0x + 0y + 0z = 2$, and no choice of x, y, z makes 0 equal 2, so the

three planes never meet in a common point.

Final answer: no solution, the system is inconsistent

Problem 4

(a) Let x_1 be the mass of Alloy 1 used, in kg; x_2 the mass of Alloy 2, in kg; and x_3 the

mass of Alloy 3, in kg. The product weighs 100 kg and is 36% copper, 44% zinc, and

20% nickel, so it holds 36 kg copper, 44 kg zinc, and 20 kg nickel. Nothing is lost in

blending, so for each metal the mass going in equals the mass coming out:

$$\text{copper: } 0.5x_1 + 0.2x_2 + 0.4x_3 = 36$$

$$\text{zinc: } 0.3x_1 + 0.7x_2 + 0.2x_3 = 44$$

$$\text{nickel: } 0.2x_1 + 0.1x_2 + 0.4x_3 = 20$$

(b) Multiplying every equation by 10 clears the decimals and keeps the elimination tidy:

$$[5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$(c) R_1 \rightarrow R_1 - 2R_3: \quad [1 \ 0 \ -4 \mid -40]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$R_2 \rightarrow R_2 - 3R_1, R_3 \rightarrow R_3 - 2R_1: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 7 \ 14 \mid 560]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R_2 \rightarrow (1/7)R_2: \quad [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R_3 \rightarrow R_3 - R_2: \quad [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 10 \mid 200]$$

$$R_3 \rightarrow (1/10)R_3: \begin{bmatrix} 1 & 0 & -4 & | & -40 \\ 0 & 1 & 2 & | & 80 \\ 0 & 0 & 1 & | & 20 \end{bmatrix}$$

Back substituting: $x_3 = 20$, then $x_2 = 80 - 2(20) = 40$, then $x_1 = -40 + 4(20) = 40$.

As a check the masses add to $40 + 40 + 20 = 100$ kg, and the copper works out to $0.5(40) + 0.2(40) + 0.4(20) = 20 + 8 + 8 = 36$ kg, exactly the target.

Final answer: 40 kg of Alloy 1, 40 kg of Alloy 2, and 20 kg of Alloy 3

Problem 5

$$(a) R_2 \rightarrow R_2 - 2R_1: \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 3 & 6 & 4 \end{bmatrix} \quad R_3 \rightarrow R_3 - 3R_1: \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2: \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad R_1 \rightarrow R_1 - R_2: \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Final answer: $\text{RREF}(A) = [1 \ 2 \ 0; 0 \ 0 \ 1; 0 \ 0 \ 0]$

(b) The reduced form has leading ones in column 1 and column 3 and a row of zeros at the

bottom, so there are exactly two pivots. Final answer: $\text{rank}(A) = 2$

(c) Running the same operations on $[A \mid b]$ turns the bottom row into $[0 \ 0 \ 0 \mid b_3 - b_1 - b_2]$,

and that one entry decides everything.

Zero solutions is possible: choosing $b = [0, 0, 1]^T$ makes the bottom row read $0 = 1$,

a contradiction, so the system is inconsistent.

Exactly one solution is impossible: column 2 is not a pivot column, so x_2 is a free variable in every consistent case, and a free variable gives a whole family of answers.

Put differently, $\text{rank}(A) = 2$ is less than the 3 unknowns, so the solution set is never

a single point.

Infinitely many is possible: if $b_3 = b_1 + b_2$, say $b = [1, 3, 4]^T$ or $b = 0$, the bottom

row reads $0 = 0$, the system is consistent, and free x_2 gives infinitely many solutions.

Problem 6

(a) I put the vectors in the columns of a matrix and row reduce, since the pivot count tells me whether they are independent.

$$\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_1: \begin{bmatrix} 1 & 2 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 + R_2: \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & -1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 2R_2: \begin{bmatrix} 1 & 0 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Two pivots for three vectors, so one of them is redundant and the set is dependent.

In fact $v_3 = v_2 - 2v_1$, since $[2, 1, 3]^T - [2, 0, 4]^T = [0, 1, -1]^T$. Because v_1 and v_2

are independent of each other, the span they fill is two dimensional.

Final answer: linearly dependent, and the span is a plane through the origin

(b) $[1 \ 0 \ 1]$

$[1 \ 1 \ 0]$

$[0 \ 1 \ 1]$

$R_2 \rightarrow R_2 - R_1: [1 \ 0 \ 1] \quad R_3 \rightarrow R_3 - R_2: [1 \ 0 \ 1]$

$[0 \ 1 \ -1] \quad [0 \ 1 \ -1]$

$[0 \ 1 \ 1] \quad [0 \ 0 \ 2]$

All three columns hold a pivot, so the only way to get $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ is

$c_1 = c_2 = c_3 = 0$, which is exactly what linear independence means. Three independent

vectors in $\mathbb{R}^3$ span the whole space.

Final answer: linearly independent, and the span is all of $\mathbb{R}^3$